Word Embeddings: Word2Vec, GloVe, FastText

Lecture 8: The Neural Revolution in NLP

PSYC 51.07: Models of Language and Communication - Week 3

Winter 2026

Today's Lecture 📋

1. 🚀 **The 2013 Revolution: Word2Vec**
2. 🏗️ **Word2Vec Architectures: CBOW & Skip-gram**
3. 🌍 **GloVe: Global Vectors**
4. 🔤 **FastText: Subword Information**
5. 📊 **Comparison & Evaluation**
6. 💡 **Applications & Best Practices**

Goal: Understand how neural methods learn dense word representations

From Count-Based to Prediction-Based 🎯

Count-Based (LSA, LDA):

- Build co-occurrence matrix
- Apply matrix factorization
- Global statistics
- Linear relationships
- Interpretable factors

Advantages:

- Fast training
- Leverages global statistics
- Mathematically grounded

From Count-Based to Prediction-Based 🎯

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Prediction-Based (Word2Vec):

- Predict context from word
- Train neural network
- Local context windows
- Non-linear relationships
- Learned representations

Advantages:

- Better semantic capture
- Scalable to huge corpora
- Dense, low-dimensional

2013: Word2Vec showed prediction-based methods could outperform count-based!

Word2Vec: The Revolution 🚀

2013: Everything changed

Key Innovation:

- Shallow neural network
- Predict context from word (CBOW)
- Predict word from context (Skip-gram)
- Dense, low-dimensional vectors
- Captures semantic relationships
- Fast training with negative sampling

Impact:

- Sparked deep learning in NLP
- Pre-training became standard
- Enabled transfer learning

Famous Results:

$$\vec{king} - \vec{man} + \vec{woman} \approx \vec{queen}$$

Other Examples:

- $\vec{Paris} - \vec{France} + \vec{Italy} \approx \vec{Rome}$
- $\vec{walking} - \vec{walk} + \vec{swim} \approx \vec{swimming}$
- $\vec{bigger} - \vec{big} + \vec{small} \approx \vec{smaller}$

Vector arithmetic captures semantic relationships!

Reference: Mikolov et al. (2013a). "Efficient Estimation of Word Representations in Vector Space"

Word2Vec: Core Intuition 💡

Words that appear in similar contexts should have similar representations

Example Context Window

"The quick brown **[TARGET]** jumped over the lazy dog"

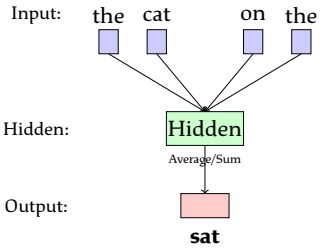
Context: [The, quick, brown, jumped, over, the, lazy, dog]
Target: fox

Key Idea:

- Train a model to predict target from context (or vice versa)
- The learned hidden layer weights become our word vectors!
- Similar words will have similar weight patterns

CBOW: Continuous Bag of Words

Predict the center word from context words



Architecture:

1. Input: One-hot vectors of context words
2. Hidden: Average of input embeddings
3. Output: Softmax over vocabulary

Training:

- Context window size (e.g., 5 words)
- Maximize $P(\text{center}|\text{context})$
- Backpropagation updates embeddings

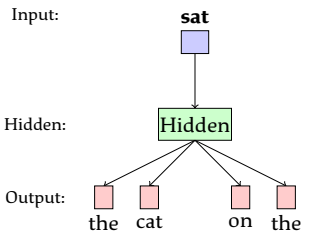
Characteristics:

- Faster to train than Skip-gram
- Better for frequent words
- Smooths over context
- Good for syntactic tasks

Objective: $\max \frac{1}{T} \sum_{t=1}^{T} \log P(w_t|w_{t-c}, \ldots, w_{t+c})$

Skip-gram: The Inverse Approach 🔄

Predict context words from the center word



Architecture:

1. Input: One-hot vector of center word
2. Hidden: Word embedding (lookup)
3. Output: Softmax over vocabulary (×C times)

Training:

- For each context position
- Maximize $P(\text{context}|\text{center})$
- Independent predictions

Characteristics:

- Slower than CBOW
- Better for rare words
- More training examples per word
- Better semantic representations

Objective: $\max \frac{1}{T} \sum_{t=1}^{T} \sum_{-c \leq j \leq c, j \neq 0} \log P(w_{t+j}|w_t)$

Negative Sampling: The Speed Trick ⚡

Problem: Softmax over entire vocabulary is too expensive!

$$P(w_O|w_I) = \frac{\exp(v_{w_O}^T v_{w_I})}{\sum_{w=1}^{V} \exp(v_w^T v_{w_I})}$$

For 100k vocabulary: Need to compute 100k exponentials per training example! 😱

Negative Sampling: The Speed Trick ⚡

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Solution: Negative Sampling

Instead of predicting across all words, create a binary classification:

- **Positive sample:** Actual context word (label = 1)
- **Negative samples:** K random words (label = 0)

$$\log \sigma(v_{w_O}^T v_{w_I}) + \sum_{i=1}^{k} \mathbb{E}_{w_i \sim P_n(w)} [\log \sigma(-v_{w_i}^T v_{w_I})]$$

Word2Vec in Practice 💻

```python
from gensim.models import Word2Vec
import gensim.downloader as api

# Load pre-trained model (300 dimensions, 3B words)
model = api.load('word2vec-google-news-300')

# Find similar words
similar = model.most_similar('computer', topn=5)
print(similar)
# Output: [('computers', 0.72), ('laptop', 0.69), ('PC', 0.68),
    ...]

# Word analogies: king - man + woman = ?
result = model.most_similar(
    positive=['king', 'woman'],
    negative=['man'],
    topn=1
```

GloVe: Global Vectors for Word Representation

Combining the best of count-based and prediction-based methods

Motivation:

- Word2Vec uses local context windows
- LSA uses global co-occurrence statistics
- Can we get the best of both?

Key Insight:
Ratios of co-occurrence probabilities encode meaning better than raw probabilities!

$P(k\|w)$	ice	steam

Ratio Encodes Relevance:

$$\frac{P(\text{solid}|\text{ice})}{P(\text{solid}|\text{steam})} \gg 1$$

$$\frac{P(\text{gas}|\text{ice})}{P(\text{gas}|\text{steam})} \ll 1$$

$$\frac{P(\text{water}|\text{ice})}{P(\text{water}|\text{steam})} \approx 1$$

Idea: Learn word vectors such that their dot product relates to co-occurrence probability ratios

GloVe: The Objective Function 🎯

Goal: Learn vectors that capture co-occurrence statistics

Objective Function:

$$J = \sum_{i,j=1}^{V} f(X_{ij}) \left(\vec{w}_i^T \tilde{\vec{w}}_j + b_i + \tilde{b}_j - \log X_{ij} \right)^2$$

where:

- X_{ij} = number of times word j appears in context of word i
- $\vec{w}_i$ = word vector for word i
- $\tilde{\vec{w}}_j$ = separate context vector for word j
- $b_i, \tilde{b}_j$ = bias terms
- $f(X_{ij})$ = weighting function

Weighting Function:

$$f(x) = \begin{cases} (x/x_{\max})^{3/4} & \text{if } x < x_{\max} \\ 1 & \text{if } x \geq x_{\max} \end{cases}$$

GloVe vs. Word2Vec 🥊

Word2Vec (Skip-gram):

- Local context windows
- Online training
- Predicts context from word
- Stochastic updates
- Negative sampling trick
- Each occurrence matters

Advantages:

- Works well on small corpora
- Can train incrementally
- Captures local patterns

GloVe:

- Global co-occurrence matrix
- Batch training
- Factorizes co-occurrence matrix
- Deterministic
- Weighted least squares
- Aggregates all occurrences

Advantages:

- Faster training (large corpora)
- Better statistical efficiency
- More interpretable

Performance

In practice: Similar performance on most tasks! Choose based on:

- Corpus size (GloVe better for large)
- Training infrastructure (GloVe needs memory for matrix)
- Availability of pre-trained models

GloVe in Practice 💻

```python
import gensim.downloader as api
import numpy as np

# Load pre-trained GloVe embeddings
# Options: glove-wiki-gigaword-50, -100, -200, -300
# Or: glove-twitter-25, -50, -100, -200
glove = api.load("glove-wiki-gigaword-100")

# Use just like Word2Vec
similar = glove.most_similar('computer', topn=5)
print(similar)

# Analogies
result = glove.most_similar(
    positive=['france', 'berlin'],
    negative=['paris'],
    topn=1
```

FastText: Enriching with Subword Information

The Problem with Word2Vec and GloVe:

Out-of-Vocabulary (OOV) Problem

- Word2Vec/GloVe: One vector per word
- New word? No vector!
- Misspelling? No vector!
- Rare morphological form? No vector!

Example:

- Have: ”running”, ”runner”
- Need: ”runnable” → OOV!
- But these words share morphology! (”run” + suffix)

FastText Solution: Represent words as bags of character n-grams

Reference: Bojanowski et al. (2017). ”Enriching Word Vectors with Subword Information”

FastText: Character N-grams

Key Idea: Break words into character n-grams

Example: ”where” (with n=3)

Character trigrams: <wh, whe, her, ere, re>
Plus: <where> (the whole word)
Word vector:

$$\vec{\text{where}} = \frac{1}{6}\left(\vec{\text{<wh>}} + \vec{\text{whe}} + \vec{\text{her}} + \vec{\text{ere}} + \vec{\text{re>}} + \vec{\text{<where>}}\right)$$

Advantages:

- Handles OOV words!
- Captures morphology
- Better for rare words
- Great for German, Turkish, Finnish

Example OOV:
Have trained on: ”run”, ”running”
Need vector for: ”runnable”
N-grams: <ru, run, unn, nna, nab, abl, ble, le>

FastText in Practice 💻

```python
from gensim.models import FastText
import gensim.downloader as api

# Load pre-trained FastText model
# Available in 157 languages!
fasttext_model = api.load('fasttext-wiki-news-subwords-300')

# Works for known words
vec_cat = fasttext_model['cat']

# Also works for unknown words! (OOV)
vec_unknownword = fasttext_model['unknownword']  # Still get a
    vector!

# Even works for misspellings (somewhat)
vec_misspelling = fasttext_model['computr']  # Close to "computer"
```

Embedding Methods Comparison 📊

Method	Year	Type	OOV	Speed	Best For
LSA	1990	Count	✗	Medium	IR, exploration
LDA	2003	Probabilistic	✗	Slow	Topic modeling
Word2Vec	2013	Neural (local)	✗	Fast	General NLP
GloVe	2014	Hybrid (global)	✗	Fast	Large corpora
FastText	2017	Neural + ngrams	✓	Fast	Morphology-rich

When to Use What?

- **Word2Vec (Skip-gram):** General purpose, good semantic quality, most popular
- **GloVe:** Large corpus, want deterministic training, good interpretability
- **FastText:** Morphologically rich languages, need OOV handling, many rare words
- **LSA/LDA:** Topic modeling, document similarity, interpretable dimensions

Typical dimensions: 50-300 (100-300 most common)

Evaluating Word Embeddings

Intrinsic Evaluation:

1. Word Similarity

- WordSim-353 dataset
- SimLex-999 dataset
- Spearman correlation with human judgments

2. Word Analogies

- Google Analogy Dataset (19k questions)
- Semantic: *Athens:Greece::Baghdad:Iraq*
- Syntactic: *good:better::bad:worse*
- Accuracy: % correct

3. Categorization

- Cluster words
- Compare to gold categories

Extrinsic Evaluation:
Use embeddings as features in downstream tasks:

- **Sentiment Analysis**
- **Named Entity Recognition**
- **POS Tagging**
- **Machine Translation**
- **Question Answering**
- **Text Classification**

Key Insight

Intrinsic scores don't always correlate with extrinsic performance!
Best approach: Evaluate on your actual task.

Limitations of Static Embeddings ⚠️

The fundamental problem: One vector per word type

Example: Polysemy

1. "I deposited money at the **bank**" (financial institution)
2. "We sat by the river **bank**" (riverside)
3. "The plane will **bank** left" (tilt)

All get the same vector! This conflates different meanings.

Other Limitations:

- No compositional semantics
- "hot dog" ≠ "hot" + "dog"
- Bias in embeddings
- Limited to training corpus
- No understanding of negation
- Missing multimodal grounding

The Solution:

- Contextual embeddings!
- Different vector per occurrence
- Consider sentence context
- ELMo, BERT, GPT, ...
- Coming next lecture!

Bias in Word Embeddings ⚖️

Embeddings learn the biases in their training data

Gender Bias Examples

- $\vec{\text{man}}$: $\vec{\text{programmer}}$ :: $\vec{\text{woman}}$: $\vec{\text{homemaker}}$
- $\vec{\text{father}}$: $\vec{\text{doctor}}$ :: $\vec{\text{mother}}$: $\vec{\text{nurse}}$
- ”man” more associated with ”career”, ”woman” with ”family”

Other Biases:

- Racial/ethnic stereotypes
- Age bias
- Religious bias
- Socioeconomic bias

Why This Matters:

- Embeddings used in hiring tools, search, recommendations
- Can perpetuate and amplify societal biases
- May violate fairness and anti-discrimination principles

References: Bolukbasi et al. (2016). ”Man is to Computer Programmer as Woman is to Homemaker?”
Caliskan et al. (2017). ”Semantics derived automatically from language corpora contain human-like biases”

Debiasing Approaches

How can we reduce bias in embeddings?

1. Post-processing:

- Identify bias subspace
- Neutralize: Remove bias from neutral words
- Equalize: Ensure equal distances

2. Training-time:

- Counterfactual data augmentation
- Adversarial debiasing
- Constrained optimization

3. Data curation:

- Balanced corpora
- Careful source selection
- Remove problematic content

Challenges:

- Hard to define "bias"
- May harm performance
- Bias is multidimensional
- Not a complete solution
- Ethical considerations

Important

Debiasing is an active research area. No perfect solution yet!
Best practice:

- Be aware of biases
- Document training data
- Test for bias

Practical Tips for Using Embeddings 💡

1. **Start with pre-trained models**
   - Word2Vec: Google News (300d, 3B words)
   - GloVe: Common Crawl (300d, 840B tokens)
   - FastText: Available in 157 languages
2. **Fine-tune if you have domain data**
   - Medical: PubMed, clinical notes
   - Legal: case law, contracts
   - Social media: tweets, posts
   - Can significantly improve performance
3. **Choose dimensions wisely**
   - More dims = more expressiveness, more data needed
   - 50-100d: small datasets, fast computation
   - 200-300d: large datasets, better quality
4. **Use cosine similarity**
   - $\text{similarity}(u, v) = \frac{u \cdot v}{||u|| \cdot ||v||}$
   - Not Euclidean distance (direction matters more than magnitude)
5. **Be aware of biases and limitations**
   - Test for bias in your use case
   - Static embeddings can't handle polysemy
   - Consider contextual embeddings for better performance

Real-World Applications

Search & Information Retrieval:

- Semantic search
- Query expansion
- Document ranking
- Question answering

Text Classification:

- Sentiment analysis
- Spam detection
- Topic classification
- Intent detection

Sequence Labeling:

- Named entity recognition
- POS tagging
- Chunking

Recommendation Systems:

- Product recommendations
- Content recommendations
- User similarity

Machine Translation:

- Word alignment
- Transfer learning
- Multilingual models

Creative Applications:

- Poetry generation
- Analogy discovery
- Semantic exploration
- Visualization

Example: Google Search uses embeddings to understand query intent and match relevant documents!

Discussion Question 💬

Why do word analogies work so well?

The famous example:

$$\vec{\text{king}} - \vec{\text{man}} + \vec{\text{woman}} \approx \vec{\text{queen}}$$

Think about:

- What does vector subtraction represent linguistically?
- Why does this capture semantic relationships?
- What are the limitations of this approach?
- Does this mean embeddings "understand" gender?

Deeper Question

Are these embeddings truly capturing *meaning*, or just statistical patterns?

Does the distributional hypothesis have limits?

Summary 🎯

What we learned today:

1. **Word2Vec (2013):** Neural revolution in embeddings
   - CBOW: Predict center from context
   - Skip-gram: Predict context from center
   - Negative sampling for efficiency
2. **GloVe (2014):** Combining count + prediction
   - Global co-occurrence statistics
   - Factorization objective
   - Ratio of probabilities
3. **FastText (2017):** Subword information
   - Character n-grams
   - Handles OOV words
   - Captures morphology
4. **Evaluation:** Intrinsic vs. extrinsic
5. **Limitations:** Static, biased, no polysemy
6. **Next:** Contextual embeddings solve polysemy!

Key References

Foundational Papers:

- Mikolov et al. (2013a). "Efficient Estimation of Word Representations in Vector Space"
- Mikolov et al. (2013b). "Distributed Representations of Words and Phrases and their Compositionality"
- Pennington et al. (2014). "GloVe: Global Vectors for Word Representation"
- Bojanowski et al. (2017). "Enriching Word Vectors with Subword Information"

Bias & Ethics:

- Bolukbasi et al. (2016). "Man is to Computer Programmer as Woman is to Homemaker?"
- Caliskan et al. (2017). "Semantics derived automatically from language corpora contain human-like biases"

Theory:

- Boleda, G. (2020). "Distributional Semantics and Linguistic Theory"
- Levy & Goldberg (2014). "Neural Word Embedding as Implicit Matrix Factorization"

Tools:

- Gensim: https://radimrehurek.com/gensim/
- Pre-trained vectors: https://github.com/RaRe-Technologies/gensim-data

Questions? 🙋

Next Lecture:

Contextual Embeddings: ELMo, Universal Sentence Encoder, BERT

One word, multiple meanings!